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Transfer learning on universal interatomic potential embeddings improves generalization in structure-property defect models

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Abstract

Oxygen vacancy formation energies govern the performance of metal oxides across energy conversion, catalysis, and electronics, yet predicting them accurately without density functional theory calculations for novel chemical systems remains challenging. Here, we show that frozen 256-dimensional embeddings extracted from a pretrained MACE universal machine-learning interatomic potential encode information sufficient to predict, to reasonable accuracy, vacancy formation energies obtained from density functional theory without requiring supercell construction, vacancy creation, or geometry optimization. A lightweight multilayer perceptron trained on these embeddings (MACE-dGNN) achieves an element-wise cross-validation mean absolute error of 0.38 eV, halving the error of a baseline graph neural network trained from scratch (0.72 eV) and offering a modest advantage over direct MACE relaxation calculations (0.51 eV). Because the embedding approach decouples representation from the prediction task and generalizes well with limited data, it will (1) extend to the prediction of properties (e.g., charged defects) that are inaccessible to generic energy- and force-output interatomic potentials and (2) be trainable on necessarily small datasets, potentially enabling the use of more expensive but accurate tools for generating training data (e.g., hybrid functional calculations) for which training or fine-tuning of interatomic potentials will be difficult. We demonstrate practical impact by revisiting a prior screening of thermochemical water-splitting materials, where improved generalization alters ∼45 % of candidate classifications.

Popular Summary. Foundation AI models, similar to those used for language and image generation, are now trained on the structures of millions of materials. Whether the knowledge stored in these models can be repurposed for problems they were not trained on remains an open question. We tested this for a property central to fuel cells, hydrogen-producing catalysts, electronic materials, and batteries: the energy needed to remove a single oxygen atom from a crystal. Without simulating the defect itself, a small network built on the foundation model's internal representations halved prediction errors compared with a model trained from scratch. When we re-ran a recent search for hydrogen-producing oxides with this approach, nearly half the candidates were reordered. The result suggests that the internal representations of pretrained materials models

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can be reused for properties they were never trained to predict, extending their usefulness beyond the energies and forces on which they were trained.

I. INTRODUCTION

Oxygen vacancies are among the most extensively studied point defects in metal oxides, governing performance across application areas in energy conversion, catalysis, and electronics. The energy required to form an oxygen vacancy, that is, to remove a lattice oxygen atom and accommodate the resulting structural and electronic reorganization, controls ionic conductivity in solid oxide fuel cells and electrolyzers [1–4], catalytic activity through the Mars–van Krevelen mechanism [5–8], resistive switching behavior in memristors and neuromorphic devices [9–11], redox thermodynamics in thermochemical water-splitting cycles [12], and the properties of optically active defect centers relevant to quantum information science [13–15]. The optimal vacancy formation energy differs substantially across applications, ranging from $\approx 0.5-1.5\,\mathrm{eV}$ for ionic conductors to $\approx 2-4\,\mathrm{eV}$ for thermochemical redox materials [12, 16–18], yet the chemical space of candidate metal oxides is vast. Navigating this design space requires predictive models that can rapidly screen tens of thousands of known and hypothetical oxides in databases such as the Materials Project [19, 20].

While density functional theory (DFT) calculations provide reliable vacancy formation energies, the cost of relaxing defective supercells, which generally contain hundreds of atoms, limits their throughput. Surrogate models that predict defect energetics from host-structure information alone are therefore valuable for high-throughput screening. A critical requirement for such models is out-of-sample generalization: because materials discovery tasks often target compositions, space groups, chemical systems, etc., absent from the training data [21], stricter performance metrics can be defined than accuracy on randomly held-out structures (structure-wise cross-validation, CV, in which all vacancies of a randomly selected subset of host structures are withheld from training), e.g., accuracy in element-wise CV test splits in which all structures containing a given cation are withheld from training. In prior work, we introduced a defect graph neural network (dGNN) that adapts the crystal graph convolutional neural network (CGCNN) architecture [22] to predict site-resolved vacancy formation energies from the host crystal graph [23]. Trained on ∼1,900 DFT-computed neutral vacancy energies in metal oxides (of which ∼1,100 are oxygen vacancies and the

rest cation vacancies), the dGNN achieves a structure-wise CV mean absolute error (MAE) of ≈0.38 eV for oxygen vacancies and has enabled experimental discovery of thermochemical water-splitting materials with ∼80 % true-positive rates [12]. However, the element-wise MAE roughly doubles to ≈0.75 eV [12], reflecting the fundamental difficulty of extrapolating to underrepresented or novel chemistries from a small, specialized training set.

Universal machine-learning interatomic potentials (uMLIPs) offer a potential solution. Foundational models built on the MACE architecture [24, 25], such as MACE-MP-0 [26] and the MACE-OMAT models trained on the Open Materials 2024 dataset [27], as well as CHGNet [28] and M3GNet [29], are trained on hundreds of thousands to millions of structures spanning much of the periodic table, encoding chemical knowledge that is unavailable to specialized models trained on small, curated datasets. They provide two routes to potentially improve element-wise generalization. First, *if* the target property prediction task relies only on structure inputs (atomic positions and lattice vectors) and force/energy outputs, such as the neutral vacancy formation energies studied herein, uMLIPs can be applied in zero-shot with no additional training required; this approach provides high-quality predictions for data points that are chemically and structurally in-distribution relative to the uMLIP's large training data, but would otherwise be out-of-distribution relative to small, domain-specific curated datasets needed for bespoke model training. The second alternative is to ask whether the latent space (specifically, crystal site embeddings) learned by uMLIPs across diverse structure and chemistry space can be repurposed by training low-complexity downstream models on these embeddings, and whether this could yield better element-wise out-of-distribution generalization than direct uMLIP relaxations or bespoke, trained-from-scratch models (dGNN).

Indeed, a growing body of work demonstrates that the embeddings learned by these potentials can be extracted and repurposed for downstream property prediction. The DPA-1 model enables efficient fine-tuning for defect and surface energies when pretrained on large datasets [30]. The HackNIP pipeline extracts embeddings from pretrained potentials for use in shallow machine-learning models [31], while the "franken" framework adapts MACE-MP0 atomic descriptors via kernel methods to predict surface and defect energies [32]. Per-atom embeddings from pretrained M3GNet and MACE potentials have also been used as descriptors for kernel ridge regression of nuclear magnetic resonance chemical shifts in organic molecules [33]. Δ-learning approaches on internal uMLIP representations can further cor-

rect systematic errors for challenging systems [34]. These studies establish that foundational uMLIP embeddings encode chemically meaningful, transferable information; however, their effectiveness depends on the similarity between source and target domains, and systematic errors can arise for high-energy configurations underrepresented in bulk-dominated training sets [35, 36]. Whether foundational uMLIP embeddings specifically improve predictions of oxygen vacancy formation energies, and whether such improvement manifests in the element-wise generalization regime, has not been systematically investigated.

Here, we address this question by extracting 256-dimensional embedding vectors from a pretrained MACE uMLIP [24–26] for crystallographic sites in host structures and using them as input features to predict DFT-calculated vacancy formation energies. We compare a hierarchy of structure-property, MACE-derived models, ranging from linear regression on scalar site energies (MACE-E_i), through linear regression on full embedding vectors (MACE-LR), to a multilayer perceptron (MLP) on embeddings (MACE-dGNN), against the original dGNN [23], which learns representations from scratch, and against a workflow using direct MACE relaxations to compute vacancy formation energies (MACE-rlx). We evaluate all structure-property models using nested CV with both structure-wise and element-wise data splits, focusing on the latter as the most stringent test of generalization for materials discovery. Without requiring supercell construction, vacancy creation, or geometry optimization, MACE-dGNN achieves an element-wise MAE of 0.38 eV, modestly outperforming the direct MACE relaxation workflow (0.51 eV). This result establishes that pretrained uMLIP embeddings encode sufficient information about local defect environments to support structure-property models, even for compositions or chemical systems absent from the surrogate model's training data. Simultaneously, the $\sim$50 % reduction relative to the baseline dGNN (0.72 eV) demonstrates the practical value of pretrained representations for out-of-sample generalization from small, chemically nonuniform datasets.

Because the embedding approach decouples the learned representation from the prediction target, the approach should extend to properties where direct uMLIP energy differences are inapplicable: for instance, charged-defect transition levels that depend on the Fermi level and require finite-size corrections [37–39], vacancy-migration barriers that require nudged elastic band calculations on defective supercells [40, 41], and properties requiring accuracy exceeding that of standard density functionals based on the generalized gradient approximation, e.g., the functional of Perdew, Burke, and Ernzerhof (PBE) [42]. Kiyohara et

al. [37] have already demonstrated charged-defect property predictions via a dGNN surrogate model trained from scratch. Finally, we revisit the thermochemical water-splitting screening of Ref. [12] with MACE-dGNN to assess the practical impact of improved generalization on materials discovery outcomes. While demonstrated here for neutral vacancy formation energies, the methodology should be general and extensible to other defect types, material classes, and oxide properties.

II. RESULTS

A. Vacancy formation energy training data

This paper addresses the surrogate modeling task of computing the neutral vacancy formation energy at a given atomic site in a host crystal structure. The formation enthalpy $\Delta H_{V_{\mathrm{X},i}}$ for an uncharged vacancy of species X (i.e., cation or oxygen) is defined from DFT total energies as follows [39]:

$$\Delta H^{\mathrm{DFT}}_{V_{\mathrm{X},i}} = E_{V_{\mathrm{X},i}} - E_{\mathrm{bulk}} + \mu_{\mathrm{X}}. \tag{1}$$

Here, $E_{V_{\mathrm{X},i}}$ is the DFT-computed total energy of a supercell containing the neutral vacancy $V_{\mathrm{X},i}$; E_{bulk} is the total energy of the pristine, defect-free supercell; and $\mu_{\mathrm{X}} = \mu^{\mathrm{ref}}_{\mathrm{X}} + \Delta\mu_{\mathrm{X}}$ is the elemental chemical potential of X. The reference $\mu^{\mathrm{ref}}_{\mathrm{X}}$ is the total energy of the elemental phase of the removed element (e.g., molecular O_2 for oxygen). Our existing DFT data incorporates the fitted elemental reference energies (FERE) correction scheme [43] into the reference chemical potential such that, for oxygen vacancies for example, $\mu^{\mathrm{ref}}_{\mathrm{O}} = 1/2E_{\mathrm{O_2}} + \delta\mu^{\mathrm{FERE}}_{\mathrm{O}}$, where $\delta\mu^{\mathrm{FERE}}_{\mathrm{O}} = 0.23$ eV.

The additional term $\Delta\mu_{\mathrm{X}}$ represents conditions in the specific experimental environment and must be non-positive for the host crystal structure to be thermodynamically stable relative to the elemental phase. For gas phases, it can be expressed as a function of the temperature and the partial pressure, e.g., $\Delta\mu_{\mathrm{O}} = \Delta\mu_{\mathrm{O}}(T, p_{\mathrm{O_2}})$ (see, e.g., Ref. [18]). Additional bounds on the $\Delta\mu_{\mathrm{X}}$ are placed by the thermodynamic stability of other limiting phases in the elemental phase space. Because $\Delta\mu_{\mathrm{X}}$ is a free variable not directly connected to defect calculations, for screening, we consider the case $\Delta\mu_{\mathrm{X}} = 0$ ($\mu_{\mathrm{X}} = \mu^{\mathrm{ref}}_{\mathrm{X}}$) in the following discussion except where noted.

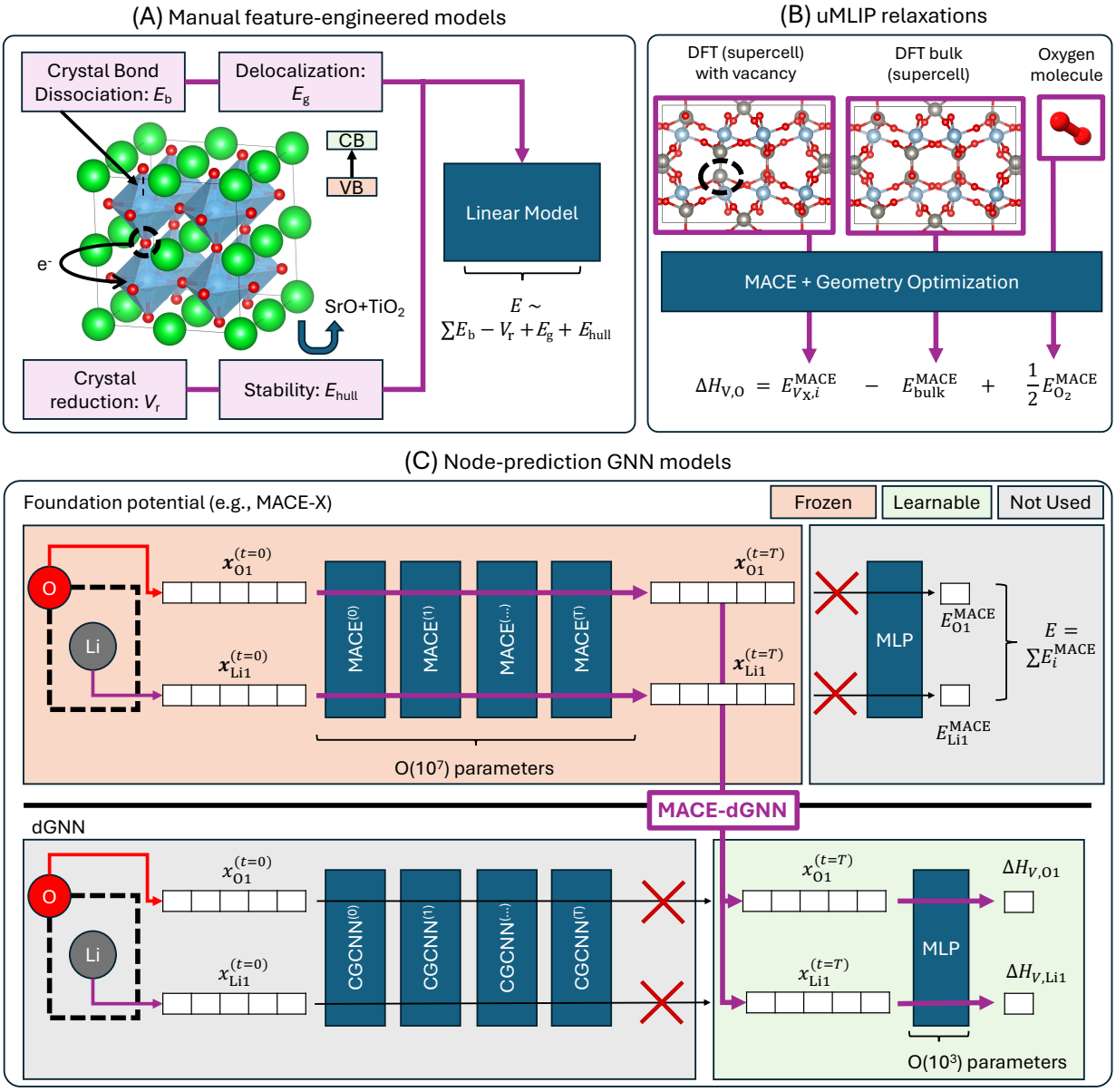


FIG. 1. Schematic examples of surrogate model alternatives to DFT for predicting point vacancy formation energies in crystalline materials. (a) Linear models using hand-selected features (E_g = band gap, E_{hull} = canonical energy above the hull, E_b = crystal bond dissociation energies, V_r = crystal reduction potential); (b) Direct use of a uMLIP (such as the MACE family of models) to perform structural relaxations needed to evaluate Equation (1). (c) A dGNN-style structure-property model that uses host structures as input (no relaxations) and uses the per-site representations produced by T frozen MACE message-passing steps MACE$^{(t=0\ldots T)}$ to generate per-site embeddings that serve as fixed inputs to a simple MLP trained for direct ΔH_{V_X} predictions.

Our $\Delta H_{V_{\mathrm{X},i}}$ training database is identical to that of Ref. [12], which merged data from Refs. [23, 44] with some additional calculations detailed therein. Specific DFT details on pseudopotentials, exchange–correlation functionals, supercell sizes, etc., can be found in Refs. [12, 23]. The database, also provided in supplementary files, contains $\Delta H_{V_{\mathrm{X},i}}$ for ∼1,900 unique crystallographic sites (of which ∼1,100 are oxygen) spanning ∼250 unique compounds. However, the coverage of elemental space in this database is not uniform, with several cations represented only by a single binary oxide structure. This imbalance, and its implications for improved model generalization, is quantified in Section II B 2 and discussed as a limitation in Section III.

B. Surrogate modeling approaches

Our first goal is to develop surrogate approaches to computing Equation (1) and compare them to the baseline dGNN [23]. The approaches investigated in this section are listed in Table I in chronological order, along with their model type and required input.

1. MACE-E_i: host site energies provide physical insights into oxygen vacancy thermodynamics

We postulate that the per-site scalar energies output by a uMLIP such as MACE (i.e., the values produced by the final linear projection applied to each site's pre-readout embedding, hereafter the "readout"; e.g., $E^{\mathrm{MACE}}_{\mathrm{bulk},i}$ in Fig. 1c) for oxygen sites in the pristine bulk should correlate with oxygen vacancy formation energies, $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{O}}}$. Here we choose the `mace-mh-0` model with the head layer trained on the Open Materials 2024 (OMAT) dataset [27] under the PBE/PBE+U exchange–correlation functional (hereafter the OMAT PBE head), to provide inference predictions most aligned with the level of DFT theory used to create our dataset. E^{MACE}_i are the per-site atomic energies that, when summed, yield the total predicted energy for a given structure: $E^{\mathrm{MACE}}_{\mathrm{tot}} = \sum_i E^{\mathrm{MACE}}_i = \sum_i \left(e^{\mathrm{MACE}}_i + \epsilon_{\mathrm{X}_i}\right)$. Thus E^{MACE}_i is the sum of e^{MACE}_i, the learned node or "interaction" energy for site i, and ϵ_{X_i}, the constant reference energy that depends only on the site's elemental identity, X_i. Figure 2a shows a positive correlation between $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{O}}}$ and $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + 1/2E^{\mathrm{MACE}}_{\mathrm{O}_2}$. A single-variable linear regression model, which we denote MACE-E_i, follows directly (though without sufficient

Model name	Brief description	Model input
dGNN	Baseline GNN model [12, 23]	DFT-relaxed host crystal structure
MACE-E_i	Single-variable linear regression model	MACE site energy
MACE-LR	256-dimensional linear regression model	MACE site embedding (after message passing)
MACE-dGNN	Two-layer (64×64) multilayer perceptron	MACE site embedding (after message passing)
MACE-sp	MACE-calculated single-point energy differences between host and vacancy-removed supercells	DFT-relaxed host crystal structure
MACE-rlx	Full MACE relaxation workflow to compute Equation (1)	MACE-relaxed host and defect supercells

TABLE I. Surrogate models used to compute vacancy formation energies in this work, listed in the order they are investigated in Section II.

accuracy):

$$\Delta H_{V_{\mathrm{O},i}}^{\text{MACE-}E_i} = 1.20\left(-E_{\mathrm{bulk},i}^{\mathrm{MACE}} + 1/2E_{\mathrm{O}_2}^{\mathrm{MACE}}\right) + 0.48. \tag{2}$$

Preceding E_i^{MACE} in the uMLIP predictions, however, is the post-message-passing, pre-readout feature vector $\boldsymbol{x}_i^{\mathrm{MACE}}$, which encodes information about each site's local environment in a low-dimensional, non-interpretable feature space. Performing a dimensionality reduction on $\boldsymbol{x}_i^{\mathrm{MACE}}$ using Uniform Manifold Approximation and Projection (UMAP) shows the following. The 2-component UMAP reduction in Figure 2b shows that cation sites tend to form relatively compact, non-overlapping clusters, while oxygen sites are non-overlapping with cation sites but are much more diffuse in the UMAP embedding space. Notably, Fe and Mn yield the most diffuse clusters among first-row transition elements, as expected based on their diversity of possible oxidation states.

Figure 2c further color-codes a 2-component UMAP reduction of just oxygen sites by $\Delta H_{V_{\mathrm{O}}}^{\mathrm{DFT}}$, revealing the local grouping of low and high vacancy formation energy sites. Thus,

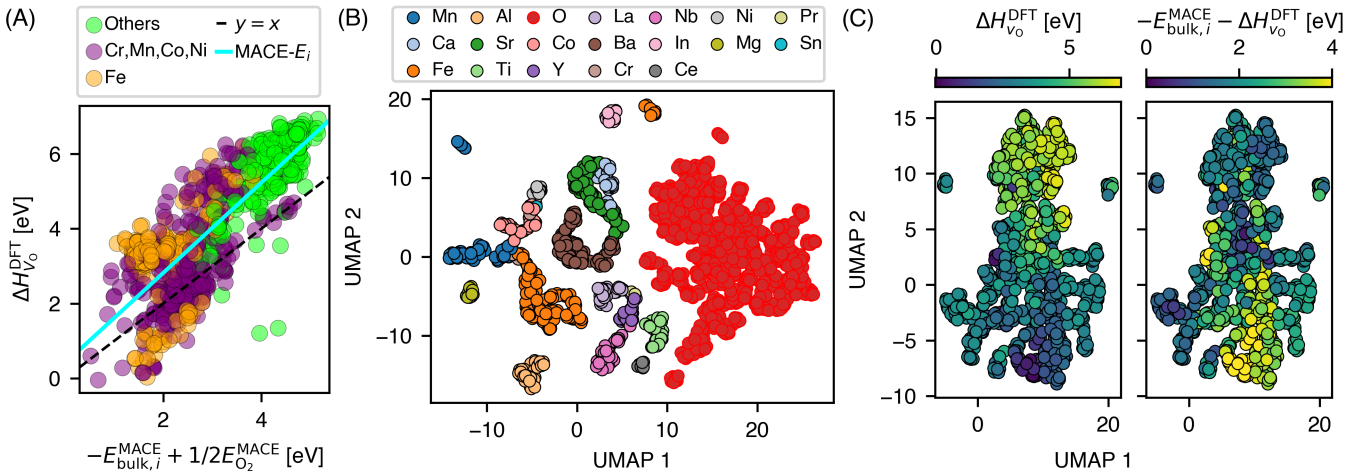


FIG. 2. (a) $\Delta H_{V_O}^{\mathrm{DFT}}$ vs. $-E_{\mathrm{bulk},i}^{\mathrm{MACE}} + 1/2E_{\mathrm{O}_2}^{\mathrm{MACE}}$, color-coded by whether their host structure contains Fe (orange), several other first-row transition metals (purple), or none of the above (green). The black line represents $y = x$ while the cyan line represents the line of best fit, Equation (2). (b) UMAP dimensionality reduction of $\boldsymbol{x}_i^{\mathrm{MACE}}$ for all vacancy sites in the database, color-coded by the elemental site identity. (c) UMAP dimensionality reduction of $\boldsymbol{x}_i^{\mathrm{MACE}}$ for all oxygen sites in the database, color-coded by $\Delta H_{V_O}^{\mathrm{DFT}}$ or $-E_i^{\mathrm{MACE}} - \Delta H_{V_O}^{\mathrm{DFT}}$.

because $-E_{\mathrm{bulk},i}^{\mathrm{MACE}}$ is less correlated to $\Delta H_{V_O}^{\mathrm{DFT}}$ for small values of $\Delta H_{V_O}^{\mathrm{DFT}}$, color-coding each site by $-E_{\mathrm{bulk},i}^{\mathrm{MACE}} - \Delta H_{V_O}^{\mathrm{DFT}}$ also shows local clustering. These results indicate that $\boldsymbol{x}_i^{\mathrm{MACE}}$ can serve as a descriptor for structure-property defect models, as examined in Section II B 2. The same embedding space can also support training-database expansion: candidate defect sites projected into the MACE embedding space that occupy sparse or unrepresented regions can be prioritized for DFT calculation, analogous to strategies that use pretrained embeddings to reduce redundant sampling in interatomic potential training [45–48].

2. MACE-LR and MACE-dGNN: Transfer learning on embeddings improves baseline predictions

We seek a structure-property surrogate model for Equation (1) that does not require manual feature engineering (in contrast to the linear or tree-based models of Refs. [38, 44, 49], schematically shown in Fig. 1a, which are generally more interpretable but less flexible). Our baseline deep learning model for structure-property defect prediction is the dGNN proposed

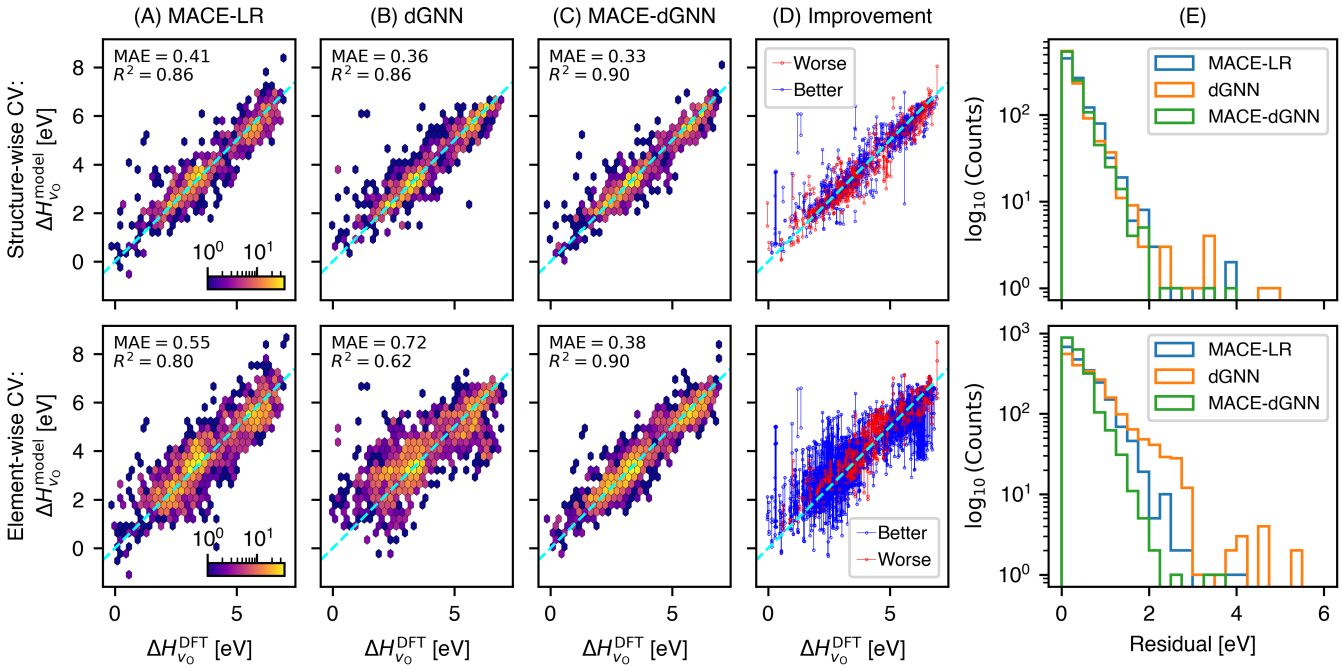


FIG. 3. Concatenation of all outer test set predictions from nested K-fold CV for the three model types considered herein: (a) MACE-LR, (b) dGNN, and (c) MACE-dGNN, for both structure-wise (top) and element-wise (bottom) hold-out strategies. Test set performance is reported only for the subset of predictions corresponding to oxygen vacancies (the full-dataset counterpart, including cation vacancies, is shown in the Supplementary Information (SI)); expectation values are computed over the outer test sets. (d) MACE-dGNN's improved (or worsened) individual test set predictions relative to dGNN. (e) Histogram of residuals for all test set predictions.

in Ref. [23] and schematically shown in Figure 1c. Trained from scratch using CGCNN convolutions followed by a readout, node-wise MLP, the dGNN predicts neutral vacancy (V_{X}) formation energies as

$$\Delta H_{V_{\text{X},i}} = f_{\text{dGNN}}(\mathcal{C}_h, i; \theta), \tag{3}$$

where $\mathcal{C}_h$ denotes the perfect (relaxed) host crystal structure, i the index of the crystallographic site hosting the vacancy, X the elemental identity of that site, and θ the learned model weights. The exact model architecture and training hyperparameters are summarized in Sec. V C. This architecture has been extended to more complex defect modeling tasks, including vacancy migration energies [40, 41] and charged vacancy formation energies [37].

Rather than training dGNN to learn local environments from scratch through its own message-passing layers, we now investigate whether the information encoded in $\boldsymbol{x}_i^{\text{MACE}}$ yields

improved predictions for $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{X},i}}$. Here we use $\boldsymbol{x}_i^{\mathrm{MACE}} \in \mathbb{R}^{256}$ from the `mace-omat-0-small` model (instead of the `mace-mh-0` model) as our input feature vector, given the small size of the dataset available for downstream training. For clarity, we summarize only test-set performance on oxygen vacancies in the main manuscript, but all downstream models were trained on the entire dataset (cation and oxygen vacancies), and CV performance across all vacancy types is summarized in the SI.

We first investigate the utility of $\boldsymbol{x}_i^{\mathrm{MACE}}$ as the input for a linear regression (LR) model, denoted MACE-LR,

$$\Delta H_{V_{\mathrm{X},i}} = \left(\boldsymbol{x}_i^{\mathrm{MACE}}\right)^\top \boldsymbol{\beta}, \tag{4}$$

where $\boldsymbol{\beta}$ are the learned linear model coefficients.

For a more expressive model that is more conceptually similar to dGNN, we use a minimal-complexity MLP for node-wise, readout predictions, denoted here as MACE-dGNN and schematically shown in Figure 1c,

$$\Delta H_{V_{\mathrm{X},i}} = \sigma_L(\boldsymbol{W}_L\sigma_{L-1}\ldots(\sigma_1(\boldsymbol{W}_1\boldsymbol{x}_i^{\mathrm{MACE}} + \boldsymbol{b}_1)\cdots + \boldsymbol{b}_{L-1}) + \boldsymbol{b}_L). \tag{5}$$

Here, $\boldsymbol{W}_l$ are learnable weight matrices, $\boldsymbol{b}$ are bias vectors, and σ_l are activation functions. More specific architecture details are given in Sec. V C, and an ablation over the hidden layers in the SI.

For each of these three models, Figure 3a–c shows the outer test set predictions from (K, L)-fold nested CV for two different criteria for allocating train/test splits (see Sec. V E for details). The top row represents $K = 10$ "Structure-wise" CV, where all defects from a given host structure are randomly assigned to the test set. The bottom row represents $K = 18$ leave-one-out "Element-wise" CV, where all defects from any host structure that contains the target test cation are assigned to the test set (with the exception of any binary oxide composition, which is always assigned to the train set). For "Structure-wise" CV, where train and test distributions much more closely overlap, the performance gain from MACE-dGNN is minimal in comparison to the baseline dGNN, although a handful of outlying errors in dGNN are improved by MACE-dGNN, as shown in Figure 3d, with the distribution of residuals specifically quantified in Figure 3e. For "Element-wise" CV, the improvement is substantially larger, with a corresponding reduction in the element-wise generalization gap.

3. *MACE-sp: single-point energy differences yield strong zero-shot performance*

We next investigate the predictive capability of MACE (specifically the `mace-mh-0` OMAT PBE head) when used only as a many-body potential energy calculator to compute a single-point vacancy formation energy (MACE-sp),

$$\Delta H_{V_{\mathrm{O},i}}^{\text{MACE-sp}} = E_{V_{\mathrm{O}},i}^{\text{MACE-sp}} - E_{\text{bulk}}^{\text{MACE-sp}} + \mu_{\mathrm{O}}^{\text{MACE}}. \tag{6}$$

Here, $E_{\text{bulk}}^{\text{MACE-sp}}$ indicates a MACE single-point total energy calculation on the *DFT-relaxed* bulk supercell (same input as dGNN, MACE-LR, and MACE-dGNN) and $E_{V_{\mathrm{O}},i}^{\text{MACE-sp}}$ indicates a MACE single-point total energy calculation on the defected supercell after removing one atom to create a vacancy but *without* relaxation. The OMAT PBE head of `mace-mh-0` was trained on raw (uncorrected) PBE/PBE+U total energies. Thus, for a consistent comparison with our FERE-corrected PBE/PBE+U data, we incorporate the same FERE correction into the MACE reference chemical potential, $\mu_{\mathrm{O}}^{\text{MACE}} = 1/2E_{\mathrm{O}_2}^{\text{MACE}} + \delta\mu_{\mathrm{O}}^{\text{FERE}}$. This yields $\mu_{\mathrm{O}}^{\text{MACE}} = -4.71\,\text{eV}$, as compared to $\mu_{\mathrm{O}}^{\text{ref}} = -4.76\,\text{eV}$ for our DFT data.

The total energies in Equation (6) can be expanded into their respective per-site energy summations,

$$\Delta H_{V_{\mathrm{O},i}}^{\text{MACE-sp}} = \sum_{j\neq i} E_{V_{\mathrm{O}},j}^{\text{MACE}} - \left(\sum_{j\neq i} E_{\text{bulk},j}^{\text{MACE}} + E_{\text{bulk},i}^{\text{MACE}}\right) + \mu_{\mathrm{O}}^{\text{MACE}}, \tag{7}$$

and rearranged to yield

$$\Delta H_{V_{\mathrm{O},i}}^{\text{MACE-sp}} = \Delta_j^{\text{MACE}} - E_{\text{bulk},i}^{\text{MACE}} + \mu_{\mathrm{O}}^{\text{MACE}}. \tag{8}$$

In this expression,

$$\Delta_j^{\text{MACE}} = \sum_{j\neq i} \delta_j^{\text{MACE}} = \sum_{j\neq i} \left(E_{V_{\mathrm{O}},j}^{\text{MACE}} - E_{\text{bulk},j}^{\text{MACE}}\right) \tag{9}$$

is the neighbor response term, i.e., the summation across all non-vacancy sites' energy differences, δ_j^{MACE}, between the pristine and defected supercell.

Figure 4a shows the MACE-sp model predictions vs. $-E_{\text{bulk},i}^{\text{MACE}} + \mu_{\mathrm{O}}^{\text{MACE}}$, such that the deviation of $\Delta H_{V_{\mathrm{O},i}}^{\text{MACE-sp}}$ from $y = x$ corresponds exactly to the Δ_j term, which is explicitly replotted in Figure 4b and color-coded by sample counts for visual clarity. While Δ_j is approximately symmetrically distributed across *all* defect sites (Fig. 4c), the behavior is

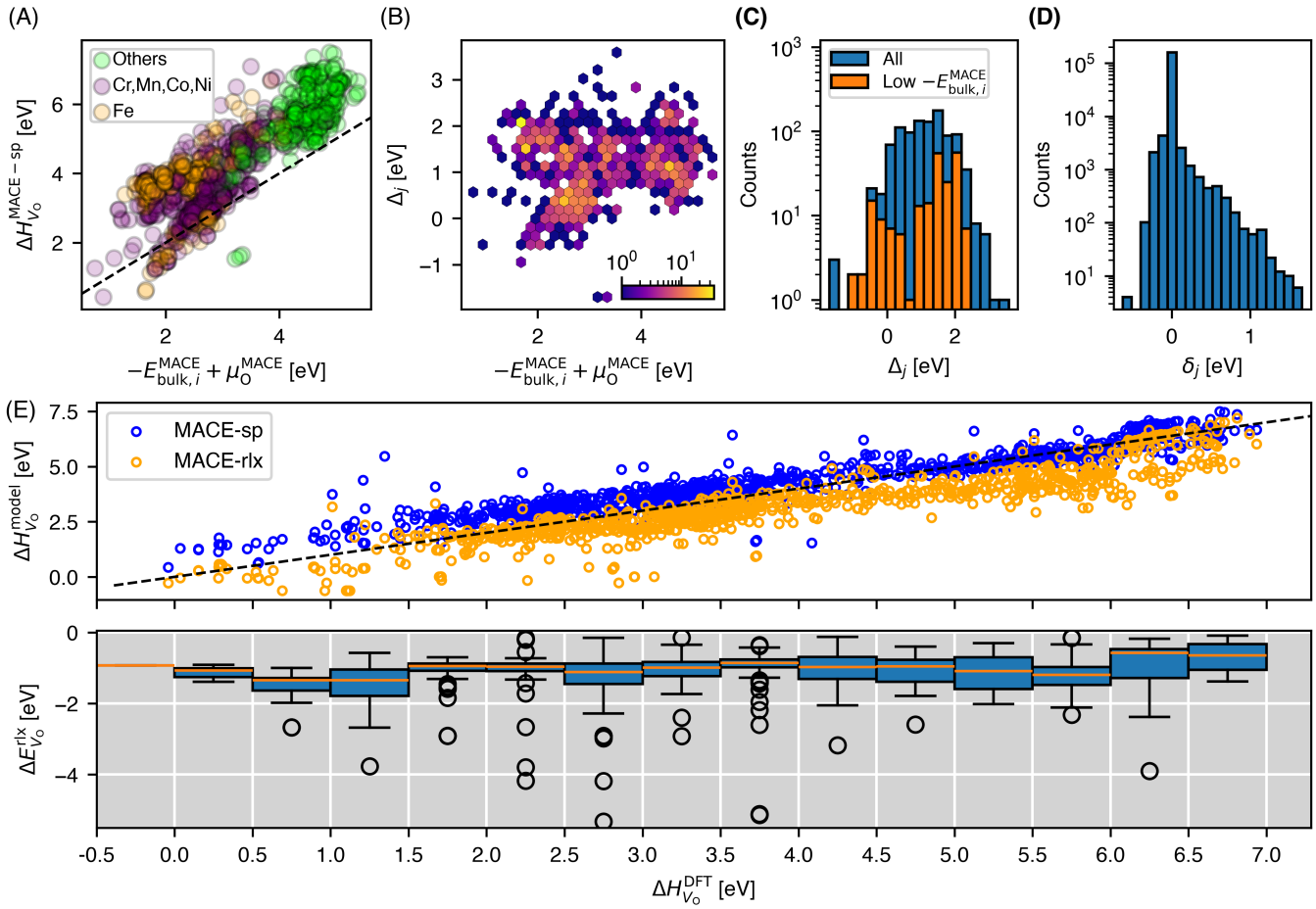


FIG. 4. (a) $\Delta H^{\text{MACE-sp}}_{V_{\mathrm{O},i}}$ vs. $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}}$, color-coded by elements present in the compound. The dashed black line represents $y = x$. (b) Δ_j vs. $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}}$, color-coded by data density (counts). (c) Distribution of Δ_j across all MACE-sp vacancy site calculations, as well as the subset where $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}} < 2.2\,\mathrm{eV}$. (d) Distribution of δ_j across all non-vacancy sites for all MACE-sp vacancy calculations. (e) Unshifted MACE-sp and MACE-rlx predictions are plotted vs. $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{O}}}$, while the boxplots beneath quantify the distribution of defect supercell relaxation energies across all defects within each $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{O}}}$ bin (outliers less than $-5.5\,\mathrm{eV}$ not visualized).

more complex for vacancies with low formation energies, $\Delta H^{\text{MACE-sp}}_{V_{\mathrm{O},i}} \lesssim 4\,\mathrm{eV}$. Here, the distribution of Δ_j broadens and becomes bimodal as $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}}$ decreases below $2.2\,\mathrm{eV}$. Finally, Figure 4d shows that, as expected, the locality of point vacancy interactions yields $\delta_j \approx 0$ for most supercell sites, and that the δ_j distribution is highly skewed with exponentially decreasing probability of large δ_j values for neighboring sites to the vacancy.

The cluster of low Δ_j vacancy sites corresponds exclusively to materials containing first-

row transition metals (Cr, Mn, Fe, Co, Ni), which exhibit variable oxidation states. For first-row transition-metal hosts, partially filled d shells permit multiple low-lying electronic configurations on the cation neighbors, so the multi-neighbor response to vacancy creation is large and varies sharply with the local environment, producing the bimodal Δ_j distribution observed at low $-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}}$. Yet the summation of weakly correlated ($-E^{\mathrm{MACE}}_{\mathrm{bulk},i} + \mu^{\mathrm{MACE}}_{\mathrm{O}}$) and Δ_j yields, via Equation (8), a raw $\Delta H^{\mathrm{MACE\text{-}sp}}_{V_{\mathrm{O}}}$ prediction (Fig. 4e) that is highly correlated with, but systematically overpredicts, $\Delta H^{\mathrm{DFT}}_{V_{\mathrm{O}}}$.

This positive bias is expected from a single-point energy model because the defected supercell is not relaxed. Solving for the constant shift that minimizes the MAE, $\mu^{\mathrm{shift-sp}}_{\mathrm{O}} = 0.36\,\mathrm{eV}$ (see Sec. V D), we denote the shifted single-point model as $\Delta H^{\mathrm{MACE\text{-}sp\text{-}}\mu}_{V_{\mathrm{O}}} = \Delta H^{\mathrm{MACE\text{-}sp}}_{V_{\mathrm{O}}} - \mu^{\mathrm{shift-sp}}_{\mathrm{O}}$, with quantitative performance summarized in Section II C. The terms $-E^{\mathrm{MACE}}_{\mathrm{bulk},i}$ and Δ_j are only weakly correlated (Fig. 4b), but MACE-sp-μ and, to a lesser extent, MACE-E_i correlate with our DFT data ($R^2 = 0.88$ and 0.66, respectively, in structure-wise CV; Table III). Together, these results indicate that MACE can (1) decompose total energies into physically meaningful per-site energies [26] and, more importantly for defect formation, (2) capture the complex many-body total energy response induced by removing an oxygen atom as the surrounding per-site energies readjust at fixed geometry.

4. MACE-rlx: direct relaxations yield strong zero-shot performance

Figure 4e shows that MACE-sp serves as a strong surrogate. Like dGNN, MACE-LR, and MACE-dGNN, however, MACE-sp requires a DFT-relaxed host structure as input. For high-throughput defect modeling coupled to hypothetical structure generation workflows, in which no DFT-relaxed host is available, we therefore also investigate whether MACE can reproduce the entire vacancy calculation workflow of Equation (1), including relaxation of the host, in direct analogy to the standard DFT approach. We compute the MACE-rlx model,

$$\Delta H^{\mathrm{MACE\text{-}rlx}}_{V_{\mathrm{O}},i} = E^{\mathrm{MACE\text{-}rlx}}_{V_{\mathrm{O}},i} - E^{\mathrm{MACE\text{-}rlx}}_{\mathrm{bulk}} + \mu^{\mathrm{MACE}}_{\mathrm{O}}, \tag{10}$$

where $E^{\mathrm{MACE\text{-}rlx}}_{\mathrm{bulk}}$ is the total energy of the pristine supercell after full MACE relaxation of cell lattice parameters and internal coordinates, and $E^{\mathrm{MACE\text{-}rlx}}_{V_{\mathrm{O}},i}$ is the total energy of the defected supercell after MACE internal coordinate-only relaxation.

Figure 4e's parity plot also shows the raw MACE-rlx predictions, which, like MACE-sp,

correlate with $\Delta H_{V_O}^{\mathrm{DFT}}$ but, unlike MACE-sp, systematically underestimate it, i.e., exhibit a large negative bias. Solving for MACE-rlx's optimal shift, $\mu_{\mathrm{O}}^{\mathrm{shift-rlx}} = -0.61$ eV, produces the MACE-rlx-μ shifted model with $\Delta H_{V_O}^{\text{MACE-rlx-}\mu} = \Delta H_{V_O}^{\text{MACE-rlx}} - \mu_{\mathrm{O}}^{\mathrm{shift-rlx}}$. This fit omits two outlier oxygen vacancy sites (see SI) with anomalously low predicted formation energies caused by instability of the defect supercell and MACE relaxation to an altogether different crystal symmetry.

Several observations are important to interpreting the relative performances of MACE-sp/MACE-rlx/MACE-sp-μ/MACE-rlx-μ vs. our DFT data. First, MACE-sp's positive bias, which yielded $\mu_{\mathrm{O}}^{\mathrm{shift-sp}} = 0.36$ eV, is unsurprising for single-point energy defect calculations where the defect state is not allowed to relax. Yet MACE-rlx's large negative bias, which yielded $\mu_{\mathrm{O}}^{\mathrm{shift-rlx}} = -0.61$ eV, is much more substantial in magnitude and beyond what can be explained by simple reference-state discrepancy arguments ($\sim$0.05 eV difference between $\mu_{\mathrm{O}}^{\mathrm{MACE}}$ and our $\mu_{\mathrm{O}}^{\mathrm{ref}}$). This negative bias indicates that MACE relaxations descend to lower energies than the corresponding DFT relaxations. Softening of the potential energy surface is a known systematic error of universal potentials [35], but whether it causes this over-relaxation would require further investigation. To further investigate, Figure 4e also shows boxplot distributions of the defect supercell MACE relaxation energies, $\Delta E_{V_O}^{\mathrm{rlx}}$, which are widely distributed but with medians that are approximately constant with $\Delta H_{V_O}^{\mathrm{DFT}}$. However, a substantial number of low-$\Delta E_{V_O}^{\mathrm{rlx}}$ outliers further drive the MACE-rlx underestimation of $\Delta H_{V_O}^{\mathrm{DFT}}$ for several vacancy sites. These tails in the $\Delta E_{V_O}^{\mathrm{rlx}}$ distribution both further skew the model towards a more negative bias and further reduce the R^2, a loss that a constant shift such as $\mu_{\mathrm{O}}^{\mathrm{shift-rlx}}$ cannot recover. This effect is pronounced enough that MACE-sp-μ maintains a slightly better performance than MACE-rlx-μ.

5. Understanding the origins of MACE-rlx's negative bias

Understanding the origins of MACE-rlx's negative bias requires detailed consideration of discrepancies in the underlying DFT settings between our data and OMAT. OMAT calculations inherit the Materials Project U corrections, which are applied only to the following elements {Co, Cr, Fe, Mn, Mo, Ni, V, W} [50]. These vary between 3.25 eV and 6.2 eV in OMAT; and, because all these elements also carry U corrections in our DFT calculations (albeit at a constant $U = 3$ eV), we denote this set as $\{U\}_{\mathrm{both}}$. Fur-

thermore, our calculations apply varying U corrections to the following set of elements, $\{U\}_{\text{dGNN}} = \{\text{Ti, Nb, Zn, La, Ce, Pr}\}$ (see SI), and these elements' U corrections are *zero* in OMAT.

Table II details the division of the MACE-rlx oxygen vacancy calculations into four mutually exclusive groups by taking the host composition's set of elements, C, and checking whether its intersection ($\cap$) with $\{U\}_{\text{dGNN}}$ and $\{U\}_{\text{both}}$ is empty ($= \emptyset$) or non-empty ($\neq \emptyset$). Figure 5 shows the distribution of the signed errors (SEs) of each group as a box plot. Unlike the global shift, the optimal per-group shifts of MACE-rlx, $\mu_{\text{O}}^{\text{shift}-\text{grp}}$ (also listed in Table II), are chemistry-specific and follow from the U coverage of the underlying OMAT and dGNN training data.

Group	$C \cap \{U\}_{\text{dGNN}}$	$C \cap \{U\}_{\text{both}}$	$\mu_{\text{O}}^{\text{shift}-\text{grp}}$ (eV)	% dataset
1	$\neq \emptyset$	$= \emptyset$	-1.29	21
2	$\neq \emptyset$	$\neq \emptyset$	-0.39	27
3	$= \emptyset$	$\neq \emptyset$	-0.55	43
4	$= \emptyset$	$= \emptyset$	-0.01	9

TABLE II. Four mutually exclusive subgroups of the MACE-rlx predictions are assigned based on whether the host composition, C, does or does not intersect with $\{U\}_{\text{dGNN}}$ and $\{U\}_{\text{both}}$. For each group, $\mu_{\text{O}}^{\text{shift}-\text{grp}}$ is the median signed error of its MACE-rlx predictions, the shift that minimizes the group's MAE (Sec. V D); the last column gives the group's percentage of the dataset.

Group 1 has a more negative bias than the dataset as a whole: its median SE is $-1.29\,\text{eV}$, compared with the global $\mu_{\text{O}}^{\text{shift}-\text{rlx}}$ of $-0.61\,\text{eV}$. When a structure contains elements from $\{U\}_{\text{dGNN}}$ (for which OMAT's $U = 0$), MACE-rlx systematically underestimates vacancy formation energy (over-relaxes the defect state) by much more than the dataset average. This is slightly mitigated for structures containing a $\{U\}_{\text{both}}$ element (Groups 2 and 3), whose mean and median SEs are closer to the dataset average, although outlier over-relaxations are still common. Group 4, whose hosts contain no elements with U corrections from either dataset, exhibits the least bias and a median SE very close to zero. However, the high frequency of over-relaxations skews the distribution and lowers the mean SE to $-0.25\,\text{eV}$. Notably, if each group's predictions are shifted by its optimal $\mu_{\text{O}}^{\text{shift}-\text{grp}}$, the overall MAE can be reduced to $0.39\,\text{eV}$, a $22\,\%$ reduction compared to globally shifted MACE-rlx-μ's

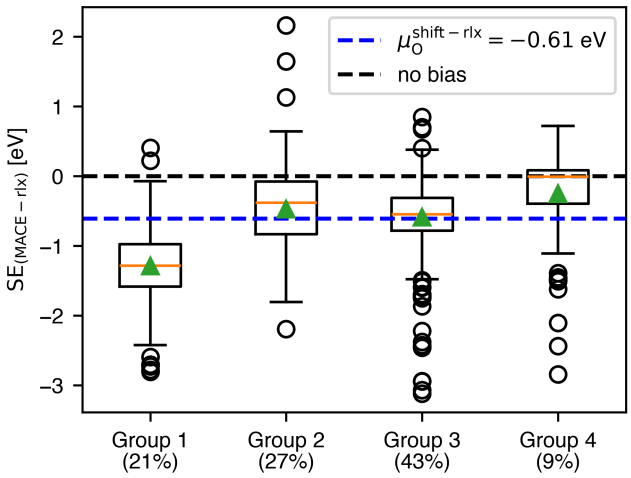


FIG. 5. Box-and-whisker plots of the signed errors (SEs) of MACE-rlx for the four groups of Table II. Orange lines mark the medians, which are the group shifts $\mu_O^{shift-grp}$, and green triangles mark the means; the blue dashed line is the global shift $\mu_O^{shift-rlx}$, and the black dashed line marks zero error. Percentages below each group label give that group's share of the oxygen vacancy dataset.

MAE of 0.50 eV (see Table III), comparable to the MAE of MACE-sp-μ (0.41 eV). With the group shifts applied, the MAE of MACE-rlx for the Group 4 hosts (also 0.39 eV) remains larger than that of MACE-dGNN for the same hosts, whether MACE-dGNN is evaluated by structure-wise (0.22 eV) or element-wise (0.33 eV) CV. With its own optimal shift for these hosts, MACE-sp has an MAE of 0.29 eV, between the structure-wise and element-wise MAEs of MACE-dGNN.

While the causes of the signed errors in Groups 1–3 cannot be separated from the differences in training U values, the negative mean SE and the low outliers of Group 4 indicate that over-relaxation of defects by MACE-rlx contributes systematically to its negative bias. Both systematic and outlier-skewed over-relaxation of vacancy defect states by MACE is unsurprising. Configurations generated by removing atoms from a host are not part of the OMAT training data [27], so the undercoordinated cation environments created by oxygen removal are extrapolative for the potential; MACE-sp evaluates these environments only at the fixed DFT geometry, whereas MACE-rlx also depends on the gradient of the MACE potential energy surface around them. In addition, DFT relaxation of a vacancy in a first-row transition-metal host is accompanied by changes in the spin and oxidation states of the

neighboring cations, which are neither explicit degrees of freedom in MACE nor sampled in its training data. Relaxed and partially relaxed defect configurations, with their DFT forces, are therefore likely critical additions to uMLIP training data for further improving defect relaxations. Nonetheless, these added complexities highlight an advantage of the MACE-dGNN approach: it uses the physics encoded in the MACE embeddings, which may be subject to some of the above limitations, while the downstream model is trained on, and therefore aligned with, one's own small, internally self-consistent DFT dataset.

C. Improving accuracy and reducing bias for out-of-distribution predictions

Figure 6 provides a comparative performance analysis across all modeling approaches for "Structure-wise" and "Element-wise" CV. Box plots in Figure 6a–b visualize the distribution of MAE and r^2 across all outer K-fold test sets, $\{\text{MAE}\}_K$ and $\{r^2\}_K$, while stars visualize each metric computed across the concatenated test sets. We show r^2 rather than R^2 because the element-wise test sets span widely varying ranges of the vacancy formation energy in our limited dataset, so several of them have a low MAE but a strongly negative R^2 only because their target range is narrow. Table III reports both metrics, and the SI compares r^2 and R^2 for the individual element-wise test sets of MACE-dGNN and MACE-rlx-μ. The MACE-E_i, MACE-sp, and MACE-rlx models are zero-shot in the sense that none of their parameters are trained on the DFT vacancy database; they generate vacancy formation energies directly from the pretrained MACE potential and a fitted scalar shift. For these zero-shot models, the box plots correspond to performance assessments on the same individual $\{\text{MAE}\}_K$ and $\{r^2\}_K$ test splits, allowing explicit comparison on the difficult-to-predict (particularly "Element-wise") splits. Across the full dataset, dGNN maintains competitive performance in "Structure-wise" CV (with MACE-dGNN slightly superior), i.e., when inference samples are relatively in-distribution (*id*) with respect to the chemical coverage of the training data.

The behavior changes substantially for "Element-wise" CV. Even the simplest model, MACE-E_i (a two-parameter linear regression), achieves performance comparable to the dGNN baseline. All other models improve substantially over dGNN, with MACE-sp, MACE-rlx, and MACE-dGNN performing best. Little distinction can be made between the relative performance of the top three models, although each shows elevated error on several test sets.

	Structure-wise CV				Element-wise CV			
Model	MAE	$\langle\{\text{MAE}\}_K\rangle$	R^2	$\langle\{R^2\}_K\rangle$	MAE	$\langle\{\text{MAE}\}_K\rangle$	R^2	$\langle\{r^2\}_K\rangle$
dGNN	0.36	0.38	0.86	0.82	0.72	0.77	0.62	0.63
MACE-E_i	0.75	0.77	0.66	0.60	0.78	0.78	0.64	0.57
MACE-LR	0.41	0.42	0.86	0.82	0.55	0.66	0.80	0.64
MACE-rlx-μ	0.50	0.51	0.83	0.79	0.51	0.59	0.83	0.70
MACE-sp-μ	0.41	0.42	0.88	0.85	0.42	0.53	0.88	**0.79**
MACE-dGNN	**0.33**	**0.35**	**0.90**	**0.87**	**0.38**	**0.47**	**0.90**	0.74

TABLE III. Summary statistics for all models' performance, where values correspond to the blue (unweighted average across all test set predictions) and orange (ensemble average metrics across K-folds) stars in Figure 6a–b. The R^2 columns and the structure-wise $\langle\{R^2\}_K\rangle$ column give the coefficient of determination, whereas Figure 6a–b shows r^2, so these values can differ from the corresponding stars. The best value in each column is shown in bold.

Figure 6c,d combines an analysis of performance for *id* vs. out-of-distribution (*ood*) chemistry space with performance for *id* vs. *ood* target property space. The histogram shows the distribution of $\Delta H_{V_\text{O}}^{\text{DFT}}$ target values above the averaged absolute error (AE) and averaged signed error (SE) of test set predictions within each $\Delta H_{V_\text{O}}^{\text{DFT}}$ bin for structure-wise and element-wise CV. All models show some performance degradation on low $\Delta H_{V_\text{O}}^{\text{DFT}}$ vacancies, but this is particularly pronounced for dGNN and especially so in element-wise CV. In contrast, MACE-dGNN exhibits lower expected AEs and expected SEs closer to zero for low and high $\Delta H_{V_\text{O}}^{\text{DFT}}$ vacancies.

D. Transfer learning's impact on applications

We investigate whether the CV improvements of MACE-dGNN translate into differences in an application-specific task. In the context of identifying novel metal oxide candidates for thermochemical water splitting (equivalently, thermochemical hydrogen production, TCH), we repeat the dGNN-based screening reported in Ref. [12] using the MACE-dGNN model. The strictness of down-selection criteria for screening metal oxides for TCH can vary, but here we adhere to two moderately strict down-selection criteria for identifying positive or

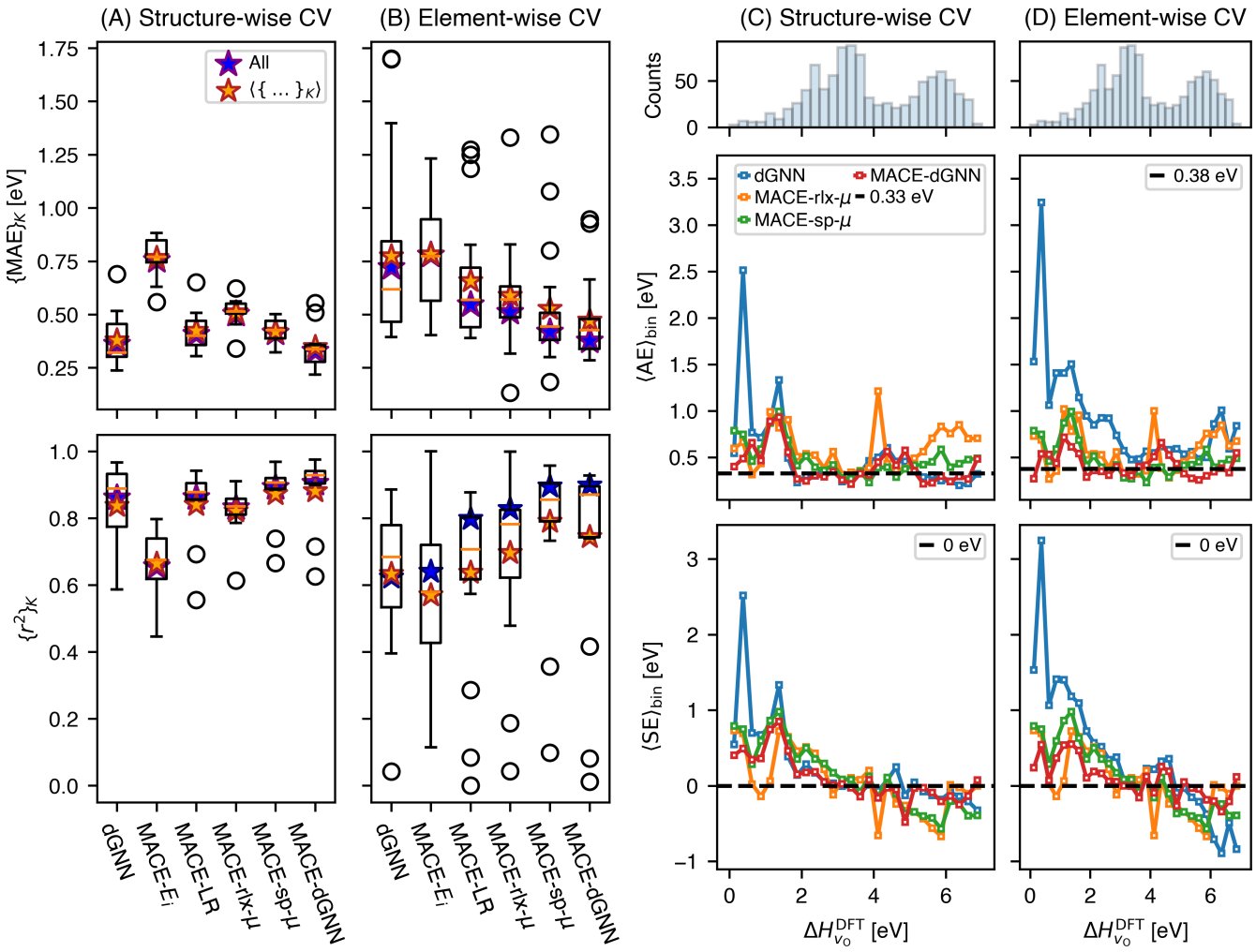


FIG. 6. Boxplot distributions of MAE and r^2 across all outer K-fold test sets, for (a) structure-wise CV and (b) element-wise CV. Blue stars represent an unweighted average of each metric across all test set predictions, while orange stars represent the ensemble average metric across K-folds. (c) Structure-wise CV and (d) Element-wise CV analysis of test set predictions showing histogram counts of the $\Delta H_{V_O}^{\mathrm{DFT}}$ target property above the averaged AE and SE of dGNN, MACE-rlx, and MACE-dGNN test set predictions within each bin. For AE plots, the dashed black line represents the MACE-dGNN global MAE, and for SE plots, the black dashed line represents 0.

screening hits.

First, our surrogate model predicts the set of all oxygen vacancy formation energies in a given material, $\{\Delta H_{V_O}\}$, the minimum of which must satisfy

$$\{\Delta H_{V_O}\}_{\min} \in [2.3, 4.0]\,\mathrm{eV}, \tag{11}$$

for a positive hit [23, 44, 51]. Based on more stringent down-selection and TCH redox operating conditions, however, even narrower target ranges have been proposed [16, 17], e.g., $[3.4, 3.9]\,\mathrm{eV}$. Second, the host structure must be stable across the typical range of oxygen chemical potentials encountered during TCH redox, i.e., $[\Delta\mu_\mathrm{O}^\mathrm{TCH}] \approx [-2.5, -3.0]\,\mathrm{eV}$ for favorable reduction and oxidation conditions [52]. As discussed when introducing Equation (1), these $\Delta\mu_\mathrm{O}$ values refer to specific experimental conditions; in the case of oxygen, they correspond to temperatures and partial pressures of O_2 gas. For each host material, we define $\phi_h(\Delta\mu_\mathrm{O})$ as the grand canonical energy above the hull as a function of the oxygen chemical potential. The range of $\Delta\mu_\mathrm{O}$ over which $\phi_h(\Delta\mu_\mathrm{O}) = 0$ (i.e., the host is on the convex hull) is denoted $[\Delta\mu_\mathrm{O}^{\phi_h=0}]$. A host is TCH-stable when this range entirely spans the operating window $[\Delta\mu_\mathrm{O}^\mathrm{TCH}]$,

$$[\Delta\mu_\mathrm{O}^\mathrm{TCH}] \in [\Delta\mu_\mathrm{O}^{\phi_h=0}]. \tag{12}$$

In practice, we require $[\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min} < -3.0\,\mathrm{eV}$ for a positive hit. Note that $[\Delta\mu_\mathrm{O}^{\phi_h=0}]$ can be computed from existing Materials Project data and its formation-energy correction schemes.

In Figure 7a–b, we show $[\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min}$ vs. $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min}$ for dGNN and MACE-dGNN screening predictions, which exhibit a negative correlation and an approximate lower bound of $-\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min} \approx [\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min}$. This lower bound is physically meaningful: below it, the true chemical-potential-dependent formation energy of oxygen vacancies, $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min} + [\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min}$, would be negative, which is unphysical and suggests that the host structure is unstable. MACE-dGNN reduces the population of outlier materials with predicted $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min} < -[\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min}$, which we interpret as a reduction in erroneous screening predictions by dGNN.

Figure 7c plots the shift between dGNN- and MACE-dGNN-predicted $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min}$, color-coded by sign and magnitude. For low (high) $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min}$ and high (low) $[\Delta\mu_\mathrm{O}^{\phi_h=0}]_\mathrm{min}$, dGNN systematically overpredicts (underpredicts) vacancy formation energies compared to MACE-dGNN. As shown in Figure 6, MACE-dGNN is more accurate (lower AE) and less biased (SE closer to zero) than dGNN across the entire target property range, particularly for *ood* materials whose compositions are underrepresented in the training data.

Figure 7d shows substantial differences between the two models when screening for negative and positive TCH hits. Of the 238 oxides for which Equations (11) and (12) are satisfied and thus identified as positives in the dGNN screening, 104 become negatives when using MACE-dGNN-predicted $\{\Delta H_{V_\mathrm{O}}\}_\mathrm{min}$. Conversely, 110 oxides that were negatives in

the dGNN screening become positives with the MACE-dGNN predictions. This reclassification arises partly because many materials have $\{\Delta H_{V_O}\}_{\min}$ near the 4.0 eV cutoff, where the dGNN and MACE-dGNN predictions can lie on opposite sides of the threshold despite differing only slightly, and partly because they differ by 1 eV or more for a substantial number of oxides. While the CV analysis in Figure 6 indicates that the MACE-dGNN predictions are more reliable, additional SI discussion of CV ensemble-based uncertainty further supports the improved performance of MACE-dGNN screening predictions.

Finally, from the screening of Ref. [12], we selected several compounds that contained cations with poor representation in the training data (Fig. 7e). We then computed their oxygen vacancy formation energies with DFT (see Methods for details), and compared them with both dGNN and MACE-dGNN predictions in Figure 7f. MACE-dGNN achieves a lower MAE than dGNN (0.42 eV vs. 0.52 eV, respectively) and correctly reclassifies one false negative as a true positive (Zn_2SnO_4) and two false positives as true negatives ($BaMoO_4$ and $FeWO_4$); however, it incorrectly reclassifies a true positive as a false negative ($FeSnO_3$). This further validates the improved generalization performance of MACE-dGNN in *ood* prediction tasks.

III. DISCUSSION

A lightweight MLP operating on frozen MACE embeddings (MACE-dGNN) achieves an element-wise generalization MAE of 0.38 eV, nearly halving the baseline dGNN error of 0.72 eV. Furthermore, MACE-dGNN's element-wise generalization modestly outperforms the zero-shot MACE calculations, MACE-rlx-μ (MAE of 0.51 eV). Following previous works that used machine-learning force fields for defect relaxations [53–56], MACE-rlx-μ replicates the DFT vacancy-energy workflow at reduced cost: defective supercells are constructed and relaxed, total-energy differences are computed, and a reference-state correction is applied. MACE-dGNN operates only on the pristine host structure: it reads the frozen embedding vector at the vacancy site and maps it to a formation energy through a shallow network, requiring neither supercell construction nor relaxation. That MACE-dGNN yields improved accuracy establishes that the pre-readout MACE embedding already encodes the local structural and chemical information relevant to vacancy formation, including the consequences of the atomic relaxation that MACE-dGNN never performs.

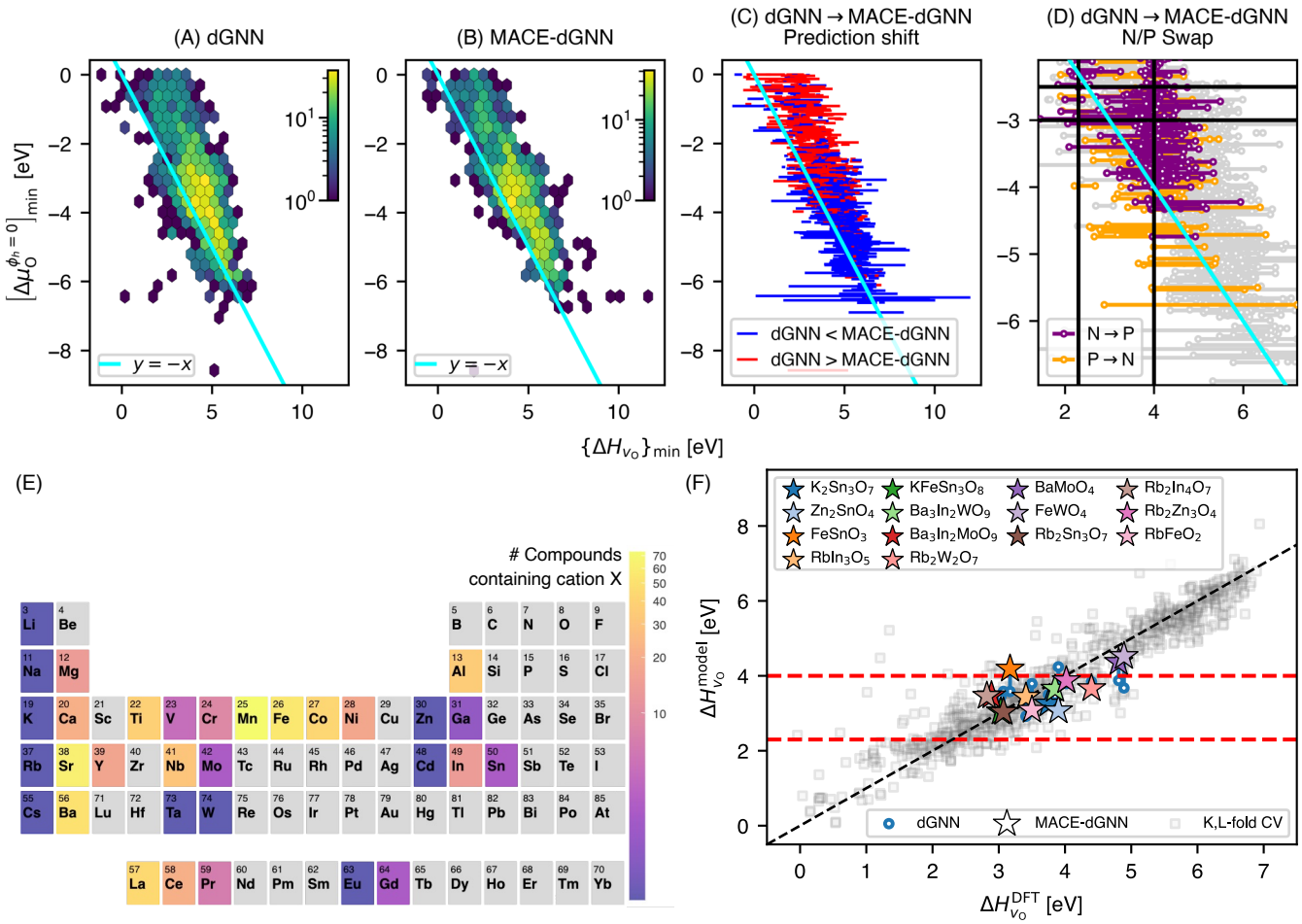


FIG. 7. Materials Project–computed $[\Delta\mu_O^{\phi_h=0}]_{min}$ for the oxide screening in Ref. [12] is plotted vs. (a) dGNN- and (b) MACE-dGNN-predicted $\{\Delta H_{V_O}\}_{min}$. The color bar corresponds to structure counts. For all subplots, the cyan line represents $y = -x$. (c) Horizontal lines connect dGNN and MACE-dGNN predictions of $\{\Delta H_{V_O}\}_{min}$ for each material, color-coded blue when the MACE-dGNN prediction is larger than the dGNN prediction and red when it is smaller. (d) Same as (c), except shifts are color-coded by whether MACE-dGNN converts a dGNN-predicted negative (N) to a positive (P) hit (purple), or vice versa (orange), for TCH down-selection; gray indicates that both models predict a negative TCH hit. The solid black lines mark the $\{\Delta H_{V_O}\}_{min}$ down-select window of $[2.3, 4.0]$ eV and the stability window $[\Delta\mu_O^{TCH}] = [-2.5, -3.0]$ eV. (e) Number of training compounds containing a given cation. (f) Original dGNN screening predictions (blue circles) for $\{\Delta H_{V_O}\}_{min}$, connected to revised MACE-dGNN predictions (colored stars), vs. the additional DFT validation calculations performed in this work. Gray squares show the structure-wise MACE-dGNN CV test predictions for comparison (i.e., data from Fig. 3c). Red dashed lines show the TCH down-selection range of $[2.3, 4.0]$ eV.

Beyond modest improvement from CV error metrics, a natural question arises as to when and why one should choose the embedding-based approach over direct uMLIP calculations. For neutral vacancy formation energies computed at the PBE level of theory, direct MACE relaxation is a viable alternative, although it still requires, as shown here, some existing DFT data on which an optimal μ^{shift} can be fit. A stronger case for the embedding approach is its potential for extensibility. Several defect properties cannot be expressed as energy differences between pristine and defective supercells within a single potential energy surface. Charged defect formation energies depend on the Fermi level position, require band-edge alignment and finite-size corrections [39, 57], and involve electronic degrees of freedom that classical interatomic potentials do not describe; recent ML approaches to this problem operate on structural features rather than total energies. Kiyohara et al. [37] have already demonstrated the ability to train dGNN-style models (i.e., learned from scratch) to predict properties related to charged defects, and we anticipate that downstream models trained on frozen uMLIP embeddings could further improve the generalizability of such models. Properties requiring beyond-PBE accuracy, such as formation energies computed with hybrid functionals [12], are accessible to the embedding approach: the MACE embeddings encode PBE-level structural chemistry that correlates with, but does not reproduce, the energetics of the Heyd–Scuseria–Ernzerhof hybrid functional, and a downstream model could in principle be trained on a small set of hybrid-functional calculations to predict such energetics directly. Related work has demonstrated this principle in periodic condensed-phase systems: machine-learning interatomic potentials for liquid water at coupled-cluster and auxiliary-field quantum Monte Carlo accuracy can be trained from as few as 200 high-level energies by initializing from a model pretrained on a cheaper density functional, whereas the same 200 energies are insufficient to train a stable model from scratch [58]. In each of these settings, the embedding approach decouples the representation (provided by the pretrained universal potential) from the prediction target (defined by the task-specific training data), enabling a single set of embeddings to serve multiple downstream tasks. Fine-tuning the potential itself on higher-fidelity data is an established alternative [58, 59], and the two routes are complementary. The distinction lies in what each requires: fine-tuning a potential requires total energies, and in practice forces, for complete configurations at the target level of theory and updates the parameters of the potential, whereas the downstream model requires only a scalar label per site, can target quantities that are not energy differences

on a single potential energy surface, and leaves the potential unchanged, so that one set of embeddings can serve several targets.

The MACE-dGNN model benefits from the chemical knowledge encoded during pretraining on millions of diverse structures, a knowledge base far larger than the ∼1,900 vacancy energies available for task-specific training. The pretrained embeddings are freely available as a community resource, and the relevant practical question is whether they improve downstream predictions for vacancy formation energies; the results presented here show that they do. A zero-hidden-layer linear regression on the 256-dimensional MACE embedding (MACE-LR) achieves an element-wise MAE of 0.55 eV, already lower than the dGNN's 0.72 eV. That a linear model on the embeddings achieves a lower MAE than the dGNN indicates that the improvement comes primarily from the quality of the pretrained representation rather than from the capacity of the regression model.

Our results contribute to a growing body of work on repurposing pretrained uMLIP representations for downstream property prediction. The "franken" framework of Novelli et al. [32] extracts MACE-MP0 atomic descriptors and could adapt them via kernel methods for surface and defect energies; the HackNIP pipeline of Kim et al. [31] similarly extracts embeddings from pretrained potentials for use in shallow ML models; and the Δ-model approach of Christiansen and Hammer [34] applies corrections to foundation model predictions using the model's own internal representations. Frozen transfer learning, in which lower-layer parameters of a pretrained MACE foundation model are held fixed during fine-tuning, has been shown to reach near-DFT accuracy for reactive surface chemistry and multi-phase alloy datasets using only 10 – 20 % of the data required to train an equivalent model from scratch [59]. Our work shares the core premise of these studies but differs in several respects. Most directly, the concurrent framework of Linton et al. [60] combines a fine-tuned CHGNet potential with a CGCNN that uses predicted Bader charges as electronic-structure descriptors to predict vacancy formation energies in Ni–Cu–Au–Pd face-centered cubic (FCC) alloys, achieving a root mean square error of 0.06 eV when training on binary and ternary data and predicting in quaternary compositions. While that workflow demonstrates the viability of bypassing DFT entirely for vacancy energetics within a fixed structure type, several distinctions are worth noting.

First, our approach uses frozen MACE embeddings with no parameter updates to the foundation model, whereas Linton et al. fine-tune CHGNet and train a separate Bader

charge predictor, creating a multi-stage workflow in which the errors of the stages can compound or, as in Δ-learning approaches, partially cancel. Because the 256-dimensional MACE embedding is the result of message passing optimized to reproduce energies and forces at scale on the OMAT dataset [27], it encodes information correlated with the local electronic environment, including features that scale with formal oxidation state and bonding character; the auxiliary Bader-charge stage of Ref. [60] supplies an analogous signal to a representation (CGCNN atom features) that does not otherwise encode it. Second, Linton et al. test generalization by compositional interpolation within the FCC lattice (binary → ternary → quaternary alloys of the same elements), whereas our element-wise CV withholds all structures containing a given cation, a more stringent test of extrapolation to novel chemistries across multiple structure types. Third, our dataset spans ∼250 oxide compounds encompassing diverse crystal structures (perovskites, spinels, rock salts, rutiles, and others), in contrast to the structurally homogeneous FCC supercells of Ref. [60]. These differences are complementary: the Linton et al. framework is suited to the data-rich, structurally uniform regime of metallic alloys, while the frozen-embedding approach demonstrated here is suited to the data-scarce, structurally diverse regime characteristic of oxide materials discovery.

Beyond uMLIP-based approaches, the most direct prior surrogate model for neutral oxygen vacancy formation energies in oxides is the random-forest model of Kumagai et al. [38], which reports MAE ≈0.34 eV using hand-engineered features on ∼1,500 oxygen vacancy calculations. On a comparable-size dataset of ∼1,100 oxygen vacancy formation energies, MACE-dGNN achieves a structure-wise MAE of 0.33 eV and an element-wise MAE of 0.38 eV (Table III) without manual feature engineering, with the latter providing an additional measurement of generalization across cation chemistry on which hand-engineered models, due to their reliance on element-specific descriptors, are typically not benchmarked. Our training data spans first-row transition metals (Mn, Fe, Co, Ni) that Kumagai et al. excluded as a methodological constraint of their PBEsol($+U$) workflow [38] but that are central to thermochemical water-splitting candidates [12]; the cation-wise stratification reported here therefore covers the chemistries most relevant to that application.

This work has several limitations. First, our training dataset encompasses ∼250 unique oxide structures with nonuniform coverage of elemental space; several cations appear in only one or two binary oxides, limiting the extent to which element-wise generalization can be assessed for those species. Second, we have tested only one family of pretrained models

(MACE models trained against OMAT data); sensitivity to the choice of foundation model, including comparisons with CHGNet, M3GNet, and models trained on other datasets, remains to be systematically characterized, as does the effect on the direct MACE workflows of larger models and of potentials that treat atomic magnetic moments as explicit degrees of freedom [61]. Third, we restrict attention to neutral vacancies; charged defect formation energies involve additional complexities (Fermi level dependence, finite-size corrections, band-edge alignment) that will require methodological extensions. Finally, the constant-shift correction applied to MACE-sp and MACE-rlx predictions uses a single global offset optimized over the full dataset. We have verified that per-fold shifts are consistent with the global value, but composition-dependent systematic errors persist after the constant correction: shifts fitted separately to the four groups of hosts defined by their overlap with $\{U\}_{\mathrm{dGNN}}$ and $\{U\}_{\mathrm{both}}$ range from $-1.29\,\mathrm{eV}$ to $-0.01\,\mathrm{eV}$ (Table II), and applying them in place of the global shift reduces the MAE of MACE-rlx from $0.50\,\mathrm{eV}$ to $0.39\,\mathrm{eV}$.

IV. CONCLUSION

We have demonstrated that frozen embeddings from a pretrained MACE universal machine-learning interatomic potential (uMLIP) encode sufficient information about local chemical environments to predict oxygen vacancy formation energies with accuracy exceeding both baseline structure-property models and direct MACE relaxation calculations. Using nested cross-validation with element-wise data splits (the most stringent benchmark for materials discovery), a lightweight multilayer perceptron on 256-dimensional MACE embeddings (MACE-dGNN) achieves a mean absolute error of $0.38\,\mathrm{eV}$, compared to $0.51\,\mathrm{eV}$ for the full MACE relaxation workflow and $0.72\,\mathrm{eV}$ for the baseline defect graph neural network (dGNN) trained from scratch. The modest advantage of the embedding-based surrogate over direct calculations establishes that pretrained uMLIP representations encode information sufficient to predict vacancy formation energies, including the effects of structural relaxation, without explicit defect calculations. Decomposing the MACE-sp formation energy into the energy of the removed site and the neighbor response term Δ_j shows that Δ_j is confined to the sites nearest the vacancy, is approximately symmetrically distributed across the dataset, and becomes bimodal for first-row transition-metal hosts, in which several low-lying electronic configurations of the neighboring cations are accessible.

Because the embedding approach separates the pretrained representation from the downstream prediction target, it should extend to properties that direct uMLIP energy differences cannot access: charged defect formation energies, vacancy migration barriers, and properties requiring levels of theory beyond the Perdew, Burke, and Ernzerhof functional. The practical significance of improved out-of-sample generalization is illustrated by a thermochemical water-splitting screening in which $\sim$45 % of candidate classifications change relative to the original dGNN predictions, showing the sensitivity of materials discovery workflows to model accuracy in the element-wise extrapolation regime.

The methodology demonstrated here is not specific to oxygen vacancies or to the MACE family of potentials. Pretrained embeddings from any uMLIP can, in principle, serve as transferable features for site-resolved property prediction, provided the foundation model's training data spans the relevant chemical and structural diversity. Systematic benchmarking across foundation models, defect types, and target properties, with consistent use of element-wise or composition-wise splits, will be needed to establish the scope and limits of this approach. As universal potentials continue to improve in accuracy and coverage, the embeddings they produce should serve as general-purpose features for materials property prediction, complementing or replacing task-specific representation learning.

V. METHODS

A. Density functional theory re-validation

To further validate MACE-dGNN screening predictions, we separately computed oxygen vacancy formation energies via density functional theory (DFT) for the fourteen compounds shown in Figure 7f, which are drawn from regions of chemical space sparsely sampled in our training dataset (Figure 7e). The generalized gradient approximation of Perdew, Burke, and Ernzerhof (PBE) was used for these calculations [42], and for compounds containing transition metals (here, Fe, Mo, W, and Zn), a Hubbard U correction of 3 eV was applied to their d orbitals. These settings follow the approach of Ref. [43] for DFT correction of thermochemical properties, which was also adopted in the original dGNN work [23]. Note that the spatially diffuse nature of Mo $4d$ and W $5d$ orbitals induces significant d-partial charges despite the formal d^0 configuration. This effect reduces the strength of the DFT+U

potential compared to the Fe $3d$ orbitals, despite the use of the same U parameter. For Zn, the training data were computed with $U = 6.0\,\text{eV}$ (see SI); the formation energies of Zn_2SnO_4 and $Rb_2Zn_3O_4$ therefore use a smaller U on Zn than the training data. The calculations used the Vienna *Ab initio* Simulation Package (VASP) [62] with the projector augmented-wave method [63, 64]. The pseudopotentials used are summarized in Table IV. We use the same FERE oxygen reference as for the training data, $\mu_O^{\text{ref}} = -4.76\,\text{eV}$, to compute these vacancy formation energies. The O_s pseudopotential allows a plane-wave cutoff energy of 360 eV. Otherwise, total energies are converged to within $10^{-6}\,\text{eV}$, and forces are converged to $10\,\text{meV}\,\text{Å}^{-1}$.

TABLE IV. Details of VASP pseudopotentials used in this work. All are taken from the VASP version 4.6 repository.

Element	POTCAR	Valence Electron Configuration
Ba	Ba_sv	$5s^2 5p^6 6s^2$
Fe	Fe	$4s^1 3d^7$
In	In_d	$5s^2 4d^{10} 5p^1$
K	K_sv	$3s^2 3p^6 4s^1$
Mo	Mo_pv	$4p^6 5s^2 4d^4$
O	O_s	$2s^2 2p^4$
Sn	Sn_d	$5s^2 4d^{10} 5p^2$
Rb	Rb_sv	$4s^2 4p^6 5s^1$
W	W_pv	$5p^6 6s^2 5d^4$
Zn	Zn	$4s^2 3d^{10}$

The compounds considered in this manner are summarized in Table V. Also listed are Materials Project identification (MP-ID) numbers, **k**-point meshes used for converging the unit cell, and supercell size, expressed as expansions along the a, b, and c lattice vectors. A single **k**-point is used for all supercell calculations: where bulk compounds are relaxed with a Γ-centered mesh, the Γ point is used; where bulk compounds are relaxed with a Monkhorst–Pack mesh, a single special **k**-point is used. For Fe-containing compounds, antiferromagnetic spin configurations minimize the total energy in all cases tested, and we maintain these spin orderings for all subsequent calculations. Spin polarization is included in all calculations,

but for compounds that do not contain Fe, no initial magnetic ordering is specified. Neutral oxygen vacancy formation energies are then calculated according to Equation (1).

TABLE V. Compounds considered for additional oxygen vacancy formation energy calculations in this work. The MP-ID numbers are provided, along with the **k**-point meshes used to converge the unit cells, and the supercell sizes used for defect calculations, expressed as expansions of the a, b, and c lattice vectors. For each **k**-point mesh, the table indicates whether it is Γ-centered or Monkhorst–Pack.

Compound	MP-ID	**k**-point mesh	Supercell size
$Ba_3In_2MoO_9$	mp-1228643	Γ, $10 \times 10 \times 6$	$2 \times 2 \times 2$
$Ba_3In_2WO_9$	mp-1228476	Monkhorst–Pack, $8 \times 4 \times 1$	$2 \times 2 \times 1$
$BaMoO_4$	mp-19276	Monkhorst–Pack, $10 \times 10 \times 2$	$2 \times 2 \times 1$
$FeSnO_3$	mp-1178212	Γ, $8 \times 8 \times 8$	$2 \times 2 \times 2$
$FeWO_4$	mp-19421	Γ, $12 \times 10 \times 12$	$3 \times 2 \times 2$
$K_2Sn_3O_7$	mp-1024073	Monkhorst–Pack, $12 \times 2 \times 1$	$3 \times 1 \times 1$
$KFeSn_3O_8$	mp-1223574	Γ, $2 \times 12 \times 4$	$2 \times 2 \times 2$
$Rb_2In_4O_7$	mp-27563	Γ, $8 \times 8 \times 2$	$2 \times 2 \times 2$
$Rb_2Sn_3O_7$	mp-2646944	Γ, $1 \times 12 \times 2$	$1 \times 3 \times 1$
$Rb_2W_2O_7$	mp-19144	Γ, $12 \times 2 \times 9$	$3 \times 1 \times 2$
$Rb_2Zn_3O_4$	mp-29606	Γ, $1 \times 6 \times 6$	$1 \times 2 \times 2$
$RbFeO_2$	mp-755536	Monkhorst–Pack, $10 \times 10 \times 6$	$2 \times 2 \times 2$
$RbIn_3O_5$	mp-1209459	Monkhorst–Pack, $8 \times 2 \times 1$	$4 \times 1 \times 1$
Zn_2SnO_4	mp-35493	Monkhorst–Pack, $8 \times 8 \times 6$	$2 \times 2 \times 2$

B. MACE embedding extraction

Embeddings were extracted from the `mace-omat-0-small` model (`https://github.com/ACEsuit/mace-foundations`), a member of the MACE family of equivariant message-passing neural networks [24, 25] pretrained on the Open Materials 2024 dataset [27] with the PBE exchange–correlation head [26]. We selected this model for two reasons: its PBE training head aligns with the level of DFT theory used in our vacancy formation energy

database, and its compact architecture ("small" variant) reduces the risk of overfitting when used to produce fixed input features for downstream models trained on our small dataset of ∼1,900 vacancy energies.

For each host crystal structure in the database, we computed a forward pass of the pretrained MACE model using the Atomic Simulation Environment interface [65]. The 256-dimensional pre-readout feature vector $\boldsymbol{x}_i^{\mathrm{MACE}} \in \mathbb{R}^{256}$ was extracted for each crystallographic site i at the output of the final MACE interaction block (T message-passing steps), before the linear energy readout that produces the scalar site energy E_i^{MACE}. These embeddings were computed once and stored; they serve as fixed (frozen) input features for all downstream models. No MACE parameters are updated during training of MACE-LR or MACE-dGNN.

C. Model architectures

We compare six models spanning a hierarchy of complexity and information access. All models predict the neutral vacancy formation enthalpy $\Delta H_{V_{\mathrm{X},i}}$ for a given crystallographic site i in a host structure.

a. MACE-E_i (scalar site energy baseline). A two-parameter linear regression on the scalar MACE site energy: $\Delta H_{V_{\mathrm{X},i}} = a \cdot E_i^{\mathrm{MACE}} + b$. This model tests whether the scalar energy readout alone correlates with vacancy formation energies.

b. MACE-LR (linear regression on embeddings). A linear regression on the full 256-dimensional embedding vector: $\Delta H_{V_{\mathrm{X},i}} = (\boldsymbol{x}_i^{\mathrm{MACE}})^\top \boldsymbol{\beta}$ (Eq. (4)).

c. Defect graph neural network (baseline, trained from scratch). The defect graph neural network (dGNN), introduced in Ref. [23], adapts the crystal graph convolutional neural network architecture [22] for site-resolved property prediction. The host crystal structure is represented as a graph with atomic nodes and distance-dependent edge features. After $T = 2$ graph convolution layers, the node feature vector corresponding to the vacancy site is extracted and passed through a fully connected readout network. The architecture comprises ∼2,000 learnable parameters, with 8-dimensional node features and 16-dimensional vacancy features. The model is trained from scratch on each cross-validation (CV) fold; full architecture and hyperparameter details are given in Ref. [23] and the supplementary information of Ref. [12].

d. MACE-dGNN (multilayer perceptron on frozen embeddings). A multilayer perceptron (MLP) operating on the frozen 256-dimensional MACE embedding of the vacancy site (Fig. 1c): $\Delta H_{V_{\mathrm{X},i}} = f_{\mathrm{MLP}}(\boldsymbol{x}_i^{\mathrm{MACE}}; \boldsymbol{\theta})$ (Eq. (5)). The MLP has two hidden layers of dimension 64 with rectified linear unit activation functions. The network is trained for a maximum of 1,000 epochs using the Adam optimizer, a learning rate of 0.001, and a batch size of 128. Training follows the same nested CV protocol and early-stopping procedure as dGNN (Sec. V E). Beyond early stopping and the averaging over inner-fold models (Sec. V E), no explicit regularization (dropout or weight penalty) is used. An ablation over the number and width of the hidden layers and over dropout regularization is reported in the SI.

e. MACE-sp and MACE-rlx (direct MACE calculations). These are zero-shot models that use MACE as an interatomic potential calculator to compute vacancy formation energies directly, as defined in Equations (6) and (10). For MACE-rlx, both the pristine and defective supercells are relaxed (cell parameters and internal coordinates for the pristine supercell, internal coordinates only for the defective supercell) using the limited-memory Broyden–Fletcher–Goldfarb–Shanno optimizer with a force convergence criterion of $0.01\,\mathrm{eV\,Å^{-1}}$. For MACE-sp, total energies are computed at the DFT-relaxed geometries without further optimization. The O_2 reference energy is taken from a MACE relaxation of the isolated molecule. A constant shift $\mu_{\mathrm{O}}^{\mathrm{shift}}$ is applied to MACE predictions to correct for the difference in reference state from the DFT data (see Sec. V D).

MACE-rlx and MACE-dGNN differ in their input requirements: MACE-rlx requires interatomic potential relaxation of both pristine bulk and defective supercells, while MACE-dGNN takes only the pristine host structure as input.

D. Direct MACE vacancy formation energy workflow

The MACE-sp and MACE-rlx workflows compute vacancy formation energies by evaluating total energy differences between pristine and defective supercells using the pretrained MACE potential, as defined in Equations (6) and (10). Because the OMAT PBE head of `mace-mh-0` was trained on total energies computed with DFT settings (elemental reference energies, energy corrections, and Hubbard U values) that differ from those of our vacancy database, the raw MACE formation energies are systematically offset from the DFT values. To remove this systematic offset, we apply a constant shift $\mu_{\mathrm{O}}^{\mathrm{shift}}$ defined as the value

minimizing the mean absolute error (MAE) of MACE predictions with respect to the DFT reference data over the full dataset:

$$\mu_{\mathrm{O}}^{\mathrm{shift}} = \arg\min_{\mu} \frac{1}{N} \sum_{i=1}^{N} \left| \Delta H_{V_{\mathrm{O},i}}^{\mathrm{MACE}} - \mu - \Delta H_{V_{\mathrm{O},i}}^{\mathrm{DFT}} \right|. \tag{13}$$

We find this shift to be $-0.61\,\mathrm{eV}$ for MACE-rlx and $0.36\,\mathrm{eV}$ for MACE-sp. To ensure that this correction does not introduce information leakage in the CV evaluation, we verified that shifts computed independently within each outer training fold are consistent with the global value. We also report unshifted MAE values in the supplementary information to enable assessment of the raw MACE prediction quality. Because $\mu_{\mathrm{O}}^{\mathrm{MACE}}$ is the same additive constant in every MACE-sp and MACE-rlx prediction, the (small) discrepancy in $\mu_{\mathrm{O}}^{\mathrm{MACE}}$ relative to the DFT reference chemical potential is absorbed exactly by $\mu_{\mathrm{O}}^{\mathrm{shift}}$; the per-element MACE reference energy ϵ_{O} likewise cancels identically between E_{bulk}, $E_{V_{\mathrm{O},i}}$, and $\mu_{\mathrm{O}}^{\mathrm{MACE}}$.

E. Cross-validation strategy

MACE-LR, dGNN, and MACE-dGNN were all trained and tested with nested (K, L)-fold CV, using identical splits across models to ensure consistency. For structure-wise splits with $(|K|, |L|) = (10, 10)$, all defects from a given structure are randomly assigned to either the outer train or test split. For element-wise splits with $(|K|, |L|) = (18, 10)$, all defects from any structure containing the targeted test cation (Al, Ba, Ca, Co, Cr, Fe, In, La, Mg, Mn, Nb, Ni, Pr, Sn, Sr, Ti, V, or Y) are held out for the outer test set (with the exception of any binary oxides, which are always assigned to the outer train set). For both outer split strategies, random $|L| = 10$-fold inner train/test splits are generated.

Taking the set of all $|L| = 10$ inner fold models' predictions on a given outer test sample, the final surrogate model's prediction is the ensemble's average,

$$\Delta H_{V_{\mathrm{X}}}^{\mathrm{model}} = 1/|L| \sum_{l \in L} \Delta H_{V_{\mathrm{X}},l}, \tag{14}$$

and the metric for uncertainty is the sample standard deviation across the L-fold ensemble,

$$\sigma(H_{V_{\mathrm{X}}}) = \sqrt{1/(L-1) \sum_{l \in L} \left| \Delta H_{V_{\mathrm{X}}}^{\mathrm{model}} - \Delta H_{V_{\mathrm{X}},l} \right|^2}. \tag{15}$$

Expected model performance for both CV splitting scenarios is assessed by MAE and the coefficient of determination (R^2, one minus the ratio of the residual to the total sum

of squares), or by the squared Pearson correlation coefficient (r^2), which, unlike R^2, does not change when all predictions are offset by a constant. Two scenarios are considered, given our dataset's imbalance (specifically in the element-wise splits), reflected in both the uneven distribution of $\Delta H_{V_X}^{\mathrm{DFT}}$ target values and the uneven coverage of chemical space. We quantify a global MAE averaged across all N outer test-set predictions in the dataset,

$$\mathrm{MAE} = 1/|N| \sum_{i \in N} \left| \Delta H_{V_X,i}^{\mathrm{model}} - \Delta H_{V_X,i}^{\mathrm{DFT}} \right|. \tag{16}$$

Alternatively, we quantify the unweighted mean of the per-fold MAEs,

$$\langle \{\mathrm{MAE}\}_K \rangle = 1/|K| \sum_{k \in K} \mathrm{MAE}_k. \tag{17}$$

The same stratification of test predictions is used to determine R^2 and $\langle \{R^2\}_K \rangle$; for element-wise CV, Table III reports $\langle \{r^2\}_K \rangle$ instead of $\langle \{R^2\}_K \rangle$ (Sec. II C). For the zero-shot models MACE-E_i, MACE-sp, and MACE-rlx, MAE, $\langle \{\mathrm{MAE}\}_K \rangle$, R^2, $\langle \{R^2\}_K \rangle$, and $\langle \{r^2\}_K \rangle$ are computed from their predictions on the same outer splits.

F. Embedding analysis

Dimensionality reduction of the 256-dimensional MACE embeddings was performed using Uniform Manifold Approximation and Projection [66, 67] as implemented in the umap-learn Python package (0.5.12). Two-component projections (Fig. 2b–c) were computed with `n_neighs_umap` = 10, `n_components` = 2, and `min_dist` = 0.99.

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DATA AVAILABILITY

The DFT vacancy formation enthalpy data on which this study is based are available at Zenodo (Ref. [68], `https://zenodo.org/records/8087871`) and in the supporting files of Ref. [44]. The training data (CIF files, defect sites, and vacancy formation energies), nested train/test splits, trained model weights, and MACE-dGNN test predictions will be provided in the supporting files upon publication. Materials Project structures used as inputs for thermochemical water-splitting screening are accessible via the Materials Project API (`https://materialsproject.org`).

CODE AVAILABILITY

The d^{2}GNN training code, which includes the dGNN model, is available at `https://github.com/sandialabs/d2gnn`. A Python script that runs the MACE-dGNN training will be provided in the supporting files upon publication. Pretrained MACE foundation models (`mace-mh-0`, `mace-omat-0-small`) are publicly available from the MACE Foundations repository (`https://github.com/ACEsuit/mace-foundations`).

AUTHOR CONTRIBUTIONS

Conceptualization: M.D.W., R.B.W. **Methodology**: M.D.W., R.B.W., S.P. **Software**: M.D.W., S.P. **Validation**: M.D.W., S.P. **Formal analysis**: M.D.W., R.B.W. **Investigation**: S.P., A.J.E.R., S.L. **Data curation**: M.D.W. **Writing – original draft**: R.B.W., M.D.W. **Writing – review and editing**: all authors. **Visualization**: M.D.W., S.P. **Supervision**: M.D.W., R.B.W., J.B.V., S.L. **Project administration**: M.D.W., R.B.W. **Funding acquisition**: M.D.W., R.B.W., J.B.V., S.L.

COMPETING INTERESTS

The authors declare no competing interests.

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